Does Pollard rho see a large prime conductor gap? A large-sample test

Study `research/conductor_gap_rho_large_sample_20261018`, rendered 2026-10-18 14:07 UTC. Unit: S = charged group operations / sqrt(r) as recorded by ecbench (`cost.s`); wall time is not read. Intervals: 95% bootstrap over targets (4000 draws) for the ratio of means; normal 95% for the paired per-target log-ratio. The host was shared (load 24–37 of 32), every run is L0.

Pooled floor / crater, arm `rho.negation`

tier / family	pairs	n floor	n crater	ratio of means	95% bootstrap	paired geo-mean	paired 95%
l	24	384	384	0.917	0.850, 0.988	0.905	0.825, 0.993
l+lr	48	768	768	0.919	0.871, 0.970	0.917	0.858, 0.979
l1	4	64	64	1.092	0.912, 1.309	1.079	0.834, 1.398
l2	4	64	64	0.927	0.778, 1.100	0.889	0.726, 1.087
l3	8	128	128	0.839	0.741, 0.951	0.851	0.725, 0.999
l4	8	128	128	0.910	0.791, 1.048	0.890	0.760, 1.041
l5r	4	64	64	0.933	0.796, 1.096	0.956	0.781, 1.170
l6r	4	64	64	0.822	0.689, 0.974	0.812	0.638, 1.033
l7r	8	128	128	0.966	0.847, 1.102	1.004	0.848, 1.189
l8r	8	128	128	0.926	0.814, 1.057	0.904	0.773, 1.057
l:cratertwin/crater	8	128	128	1.039	0.919, 1.176	1.077	0.934, 1.243
l:twin/original	8	128	128	1.039	0.919, 1.173	1.077	0.934, 1.243
lr	24	384	384	0.920	0.856, 0.993	0.928	0.846, 1.019
lr:cratertwin/crater	8	128	128	0.891	0.791, 1.005	0.878	0.745, 1.035
lr:twin/original	8	128	128	0.891	0.793, 1.006	0.878	0.745, 1.035
m	9	1152	1152	0.982	0.940, 1.027	0.994	0.941, 1.050
m1	3	384	384	1.050	0.973, 1.131	1.091	0.985, 1.208
m2	3	384	384	0.967	0.895, 1.042	0.975	0.890, 1.069
m3	3	384	384	0.933	0.865, 1.009	0.924	0.845, 1.011
s	13	13312	13312	0.998	0.986, 1.010	0.999	0.985, 1.014
s1	12	12288	12288	0.998	0.985, 1.010	1.000	0.985, 1.015
s2	1	1024	1024	1.005	0.961, 1.051	0.992	0.939, 1.048
s:cratertwin/crater	5	5120	5120	1.008	0.988, 1.028	1.008	0.985, 1.032
s:floortwin/floor	5	5120	5120	1.001	0.981, 1.022	1.000	0.976, 1.024
s:twin/original	10	10240	10240	1.004	0.991, 1.019	1.004	0.987, 1.021
t	24	384	384	0.945	0.878, 1.021	0.943	0.861, 1.034
t1	3	48	48	0.974	0.782, 1.213	0.957	0.726, 1.262
t2	3	48	48	0.994	0.813, 1.230	1.066	0.823, 1.380
t3	3	48	48	1.002	0.809, 1.257	0.986	0.778, 1.252
t4	3	48	48	0.935	0.787, 1.122	0.938	0.769, 1.144
t5	3	48	48	0.902	0.718, 1.117	0.832	0.664, 1.041
t6	3	48	48	0.979	0.788, 1.212	0.977	0.697, 1.370
t7	3	48	48	0.828	0.677, 1.026	0.846	0.655, 1.092
t8	3	48	48	0.967	0.792, 1.183	0.965	0.733, 1.271
t:cratertwin/crater	24	384	384	0.967	0.898, 1.042	0.982	0.894, 1.078
t:floortwin/cratertwin	24	384	384	1.040	0.965, 1.119	1.022	0.930, 1.122
t:floortwin/floor	24	384	384	1.064	0.987, 1.146	1.063	0.973, 1.162
t:twin/original	48	768	768	1.014	0.963, 1.066	1.022	0.958, 1.090

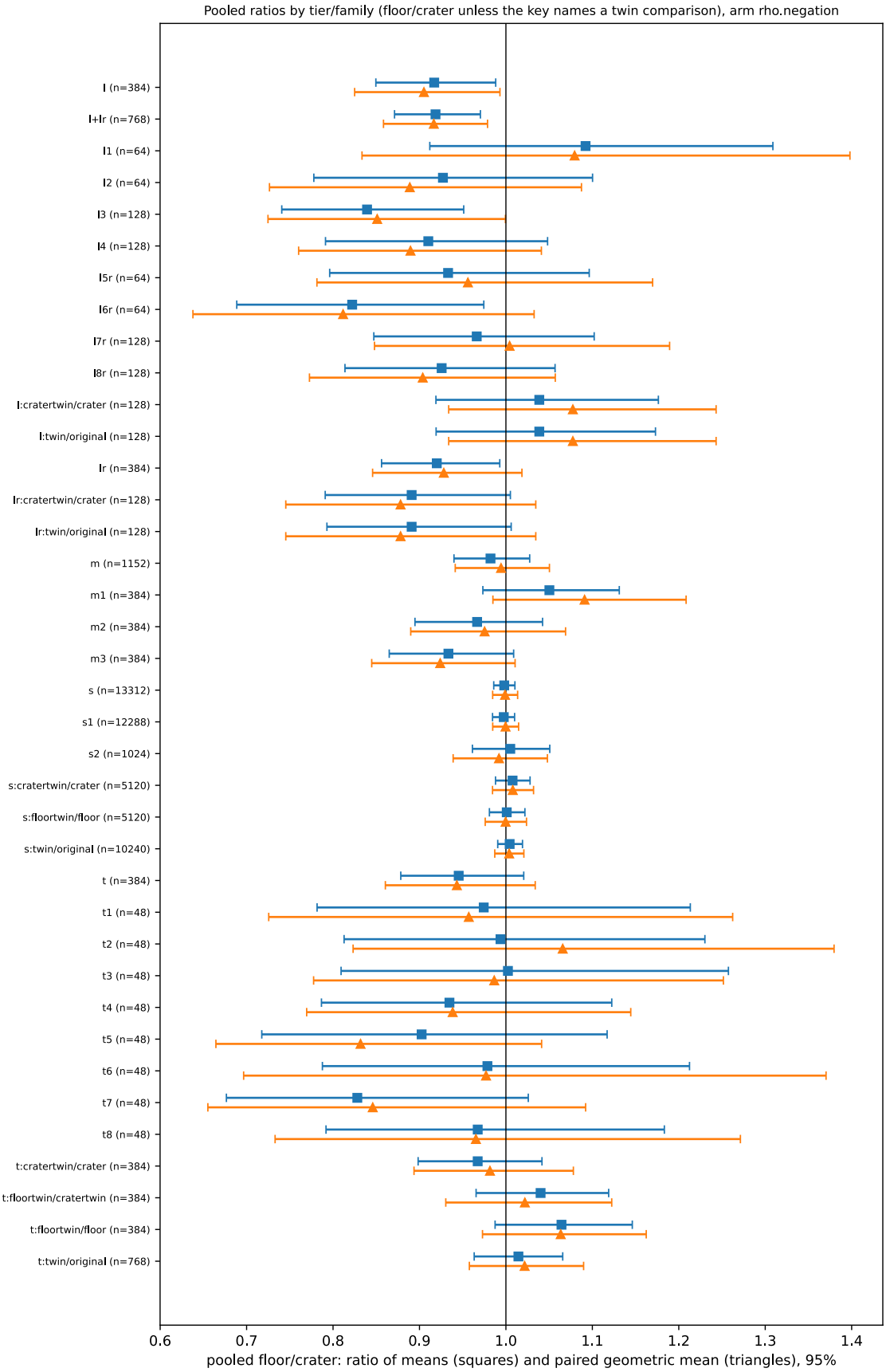


Figure 1: Pooled floor/crater by family and tier, arm rho.negation. s: ℓ = 8209 (s1a–d) and 131101 (s2); m: ℓ ≈ 2^20.2 at 45 bits; l: ℓ ≈ 2^25 at 53 bits (l1–l4 frozen, l5r–l8r replication).

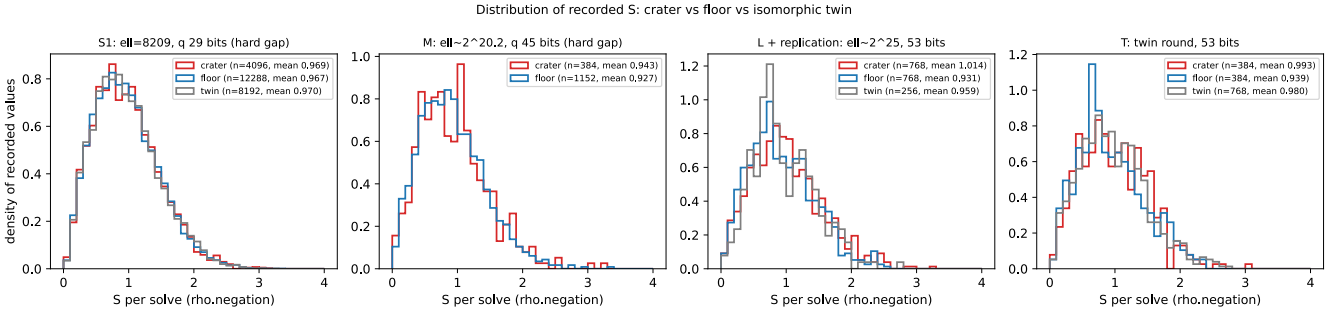


Figure 2: Recorded S per solve, crater vs floor vs isomorphic twin.

Controls

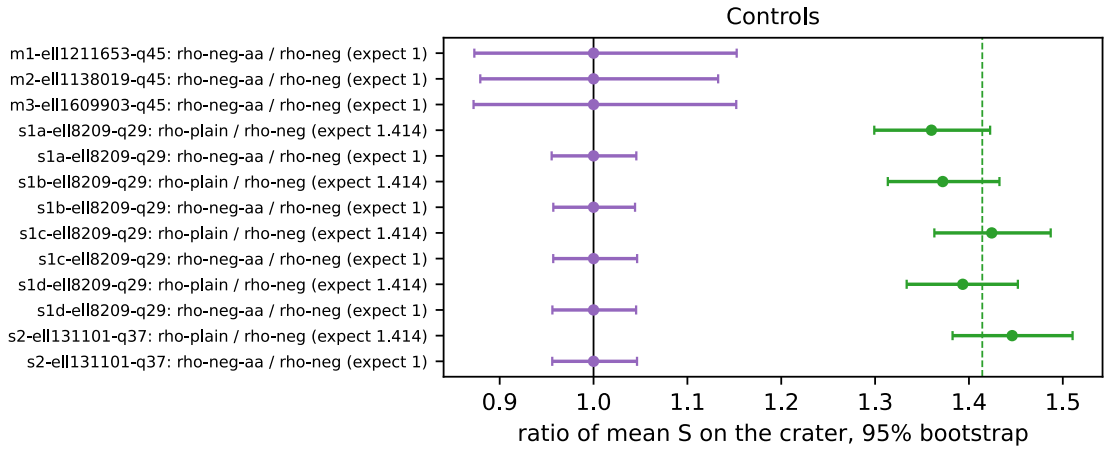


Figure 3: Positive control rho.plain / rho.negation (expected 1.414) and A/A arm (identical seeds, so exactly 1 per target).

Every comparison, arm `rho.negation`

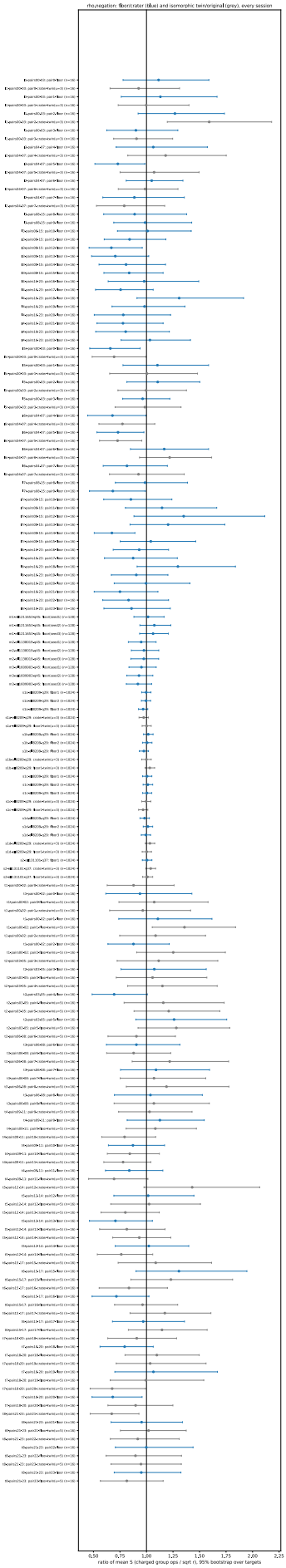


Figure 4: Every floor/crater (blue) and twin/original (grey) ratio with its 95% bootstrap interval.

l1-pairs00-03
conductor gap ell 2^25, D=-19, 53-bit, pairs 0..3 of cm_orbits: crater vs floor vs crater twins, 16 shared targets. Record outcomes: {'verified': 192}.

Arm `rho-neg`:

comparison	n	mean S base	mean S other	ratio	95% bootstrap	paired geo-mean	paired 95%
pair0-floor / pair0-crater	16/16	1.032	1.150	1.114	0.779, 1.588	1.112	0.670, 1.844
pair0-crater-twin(u=3) / pair0-crater	16/16	1.032	0.956	0.926	0.656, 1.312	0.990	0.683, 1.435
pair1-floor / pair1-crater	16/16	0.967	1.094	1.131	0.762, 1.665	1.063	0.567, 1.992
pair1-crater-twin(u=3) / pair1-crater	16/16	0.967	0.965	0.998	0.733, 1.401	1.177	0.709, 1.954
pair2-floor / pair2-crater	16/16	0.761	0.965	1.268	0.921, 1.735	1.202	0.746, 1.935
pair2-crater-twin(u=3) / pair2-crater	16/16	0.761	1.210	1.591	1.196, 2.180	1.671	1.278, 2.184
pair3-floor / pair3-crater	16/16	1.011	0.910	0.900	0.626, 1.295	0.956	0.587, 1.559
pair3-crater-twin(u=3) / pair3-crater	16/16	1.011	0.916	0.906	0.690, 1.247	1.062	0.728, 1.549

l2-pairs04-07
conductor gap ell 2^25, D=-19, 53-bit, pairs 4..7 of cm_orbits: crater vs floor vs crater twins, 16 shared targets. Record outcomes: {'verified': 192}.

Arm `rho-neg`:

comparison	n	mean S base	mean S other	ratio	95% bootstrap	paired geo-mean	paired 95%
pair4-floor / pair4-crater	16/16	0.847	0.902	1.064	0.714, 1.574	1.078	0.709, 1.638
pair4-crater-twin(u=3) / pair4-crater	16/16	0.847	1.000	1.180	0.824, 1.752	1.286	0.874, 1.892
pair5-floor / pair5-crater	16/16	0.984	0.718	0.730	0.514, 0.985	0.611	0.400, 0.933
pair5-crater-twin(u=3) / pair5-crater	16/16	0.984	1.054	1.072	0.751, 1.496	0.898	0.601, 1.342
pair6-floor / pair6-crater	16/16	0.960	1.007	1.049	0.809, 1.344	1.012	0.724, 1.415
pair6-crater-twin(u=3) / pair6-crater	16/16	0.960	0.947	0.987	0.734, 1.297	0.919	0.689, 1.226
pair7-floor / pair7-crater	16/16	0.939	0.832	0.886	0.588, 1.357	0.937	0.624, 1.407
pair7-crater-twin(u=3) / pair7-crater	16/16	0.939	0.742	0.790	0.529, 1.172	0.828	0.483, 1.420

l3-pairs08-15
conductor gap ell 2^25, D=-19, 53-bit, pairs 8..15 of cm_orbits: crater vs floor, 16 shared targets. Record outcomes: {'verified': 256}.

Arm `rho-neg`:

comparison	n	mean S base	mean S other	ratio	95% bootstrap	paired geo-mean	paired 95%
pair8-floor / pair8-crater	16/16	0.933	0.828	0.888	0.596, 1.379	0.989	0.707, 1.384
pair9-floor / pair9-crater	16/16	0.963	0.950	0.987	0.691, 1.426	1.023	0.735, 1.424
pair10-floor / pair10-crater	16/16	1.028	1.039	1.010	0.726, 1.421	1.078	0.631, 1.842
pair11-floor / pair11-crater	16/16	1.088	0.915	0.841	0.602, 1.183	0.876	0.568, 1.352
pair12-floor / pair12-crater	16/16	0.996	0.666	0.669	0.459, 0.960	0.597	0.335, 1.061
pair13-floor / pair13-crater	16/16	1.217	0.859	0.706	0.483, 1.021	0.691	0.436, 1.096
pair14-floor / pair14-crater	16/16	0.940	0.759	0.807	0.553, 1.179	0.855	0.507, 1.443
pair15-floor / pair15-crater	16/16	1.099	0.920	0.837	0.599, 1.157	0.817	0.547, 1.221

l4-pairs16-23
conductor gap ell 2^25, D=-19, 53-bit, pairs 16..23 of cm_orbits: crater vs floor, 16 shared targets. Record outcomes: {'verified': 256}.

Arm `rho-neg`:

comparison	n	mean S base	mean S other	ratio	95% bootstrap	paired geo-mean	paired 95%
pair16-floor / pair16-crater	16/16	1.095	1.071	0.978	0.639, 1.492	0.944	0.593, 1.502
pair17-floor / pair17-crater	16/16	1.148	0.868	0.756	0.520, 1.063	0.593	0.351, 1.001
pair18-floor / pair18-crater	16/16	0.772	1.010	1.308	0.910, 1.915	1.359	0.930, 1.985
pair19-floor / pair19-crater	16/16	0.904	0.887	0.982	0.675, 1.362	0.867	0.575, 1.308
pair20-floor / pair20-crater	16/16	1.031	0.806	0.781	0.508, 1.227	0.875	0.563, 1.358

pair21-floor / pair21-crater	16/16	1.201	0.935	0.778	0.534, 1.157	0.863	0.551, 1.352
pair22-floor / pair22-crater	16/16	1.011	0.813	0.804	0.522, 1.215	0.759	0.467, 1.234
pair23-floor / pair23-crater	16/16	1.059	1.094	1.032	0.760, 1.415	1.038	0.716, 1.506

l5r-pairs00-03

conductor gap ell 2^25, D=-19, 53-bit, pairs 0..3 of cm_orbits: crater vs floor vs crater twins, 16 shared targets. Record outcomes: {'verified': 192}.

Arm `rho-neg`:

comparison	n	mean S base	mean S other	ratio	95% bootstrap	paired geo-mean	paired 95%
pair0-floor / pair0-crater	16/16	1.194	0.787	0.659	0.465, 0.940	0.667	0.374, 1.189
pair0-crater-twin(u=3) / pair0-crater	16/16	1.194	0.829	0.695	0.488, 0.994	0.706	0.425, 1.172
pair1-floor / pair1-crater	16/16	0.847	0.935	1.103	0.778, 1.585	1.143	0.767, 1.702
pair1-crater-twin(u=3) / pair1-crater	16/16	0.847	0.851	1.004	0.653, 1.481	0.851	0.449, 1.616
pair2-floor / pair2-crater	16/16	0.887	0.981	1.106	0.815, 1.503	1.100	0.818, 1.479
pair2-crater-twin(u=3) / pair2-crater	16/16	0.887	0.884	0.996	0.731, 1.377	0.982	0.750, 1.286
pair3-floor / pair3-crater	16/16	0.980	0.944	0.963	0.776, 1.220	0.997	0.788, 1.263
pair3-crater-twin(u=3) / pair3-crater	16/16	0.980	0.966	0.986	0.704, 1.326	0.819	0.455, 1.472

l6r-pairs04-07

conductor gap ell 2^25, D=-19, 53-bit, pairs 4..7 of cm_orbits: crater vs floor vs crater twins, 16 shared targets. Record outcomes: {'verified': 192}.

Arm `rho-neg`:

comparison	n	mean S base	mean S other	ratio	95% bootstrap	paired geo-mean	paired 95%
pair4-floor / pair4-crater	16/16	1.207	0.820	0.679	0.445, 0.995	0.594	0.348, 1.013
pair4-crater-twin(u=3) / pair4-crater	16/16	1.207	0.934	0.774	0.552, 1.078	0.766	0.524, 1.120
pair5-floor / pair5-crater	16/16	1.426	1.043	0.732	0.533, 0.976	0.663	0.461, 0.954
pair5-crater-twin(u=3) / pair5-crater	16/16	1.426	1.037	0.727	0.557, 0.956	0.711	0.514, 0.983
pair6-floor / pair6-crater	16/16	0.893	1.042	1.167	0.849, 1.585	1.119	0.719, 1.741
pair6-crater-twin(u=3) / pair6-crater	16/16	0.893	1.088	1.218	0.934, 1.614	1.214	0.829, 1.779
pair7-floor / pair7-crater	16/16	1.034	0.843	0.816	0.592, 1.197	0.985	0.575, 1.687
pair7-crater-twin(u=3) / pair7-crater	16/16	1.034	0.956	0.925	0.651, 1.356	1.106	0.660, 1.854

l7r-pairs08-15

conductor gap ell 2^25, D=-19, 53-bit, pairs 8..15 of cm_orbits: crater vs floor, 16 shared targets. Record outcomes: {'verified': 256}.

Arm `rho-neg`:

comparison	n	mean S base	mean S other	ratio	95% bootstrap	paired geo-mean	paired 95%
pair8-floor / pair8-crater	16/16	1.005	0.991	0.987	0.706, 1.386	1.078	0.658, 1.766
pair9-floor / pair9-crater	16/16	1.098	0.750	0.683	0.463, 0.989	0.690	0.445, 1.069
pair10-floor / pair10-crater	16/16	0.778	0.663	0.853	0.597, 1.239	0.864	0.590, 1.265
pair11-floor / pair11-crater	16/16	0.703	0.806	1.147	0.801, 1.662	1.130	0.575, 2.219
pair12-floor / pair12-crater	16/16	0.822	1.111	1.350	0.882, 2.115	1.444	0.843, 2.474
pair13-floor / pair13-crater	16/16	0.897	1.080	1.204	0.845, 1.738	1.290	0.838, 1.986
pair14-floor / pair14-crater	16/16	1.201	0.809	0.674	0.508, 0.891	0.696	0.481, 1.009
pair15-floor / pair15-crater	16/16	0.994	1.034	1.041	0.752, 1.464	1.100	0.741, 1.631

l8r-pairs16-23

conductor gap ell 2^25, D=-19, 53-bit, pairs 16..23 of cm_orbits: crater vs floor, 16 shared targets. Record outcomes: {'verified': 256}.

Arm `rho-neg`:

comparison	n	mean S base	mean S other	ratio	95% bootstrap	paired geo-mean	paired 95%
pair16-floor / pair16-crater	16/16	1.057	0.984	0.931	0.686, 1.208	0.794	0.536, 1.176

pair17-floor / pair17-crater	16/16	1.192	1.042	0.874	0.604, 1.291	0.938	0.501, 1.756
pair18-floor / pair18-crater	16/16	1.059	1.373	1.296	0.912, 1.839	1.265	0.757, 2.112
pair19-floor / pair19-crater	16/16	0.955	0.863	0.904	0.669, 1.202	0.868	0.616, 1.223
pair20-floor / pair20-crater	16/16	0.943	0.939	0.996	0.696, 1.409	0.942	0.626, 1.419
pair21-floor / pair21-crater	16/16	1.023	0.767	0.750	0.507, 1.108	0.785	0.497, 1.239
pair22-floor / pair22-crater	16/16	1.339	1.112	0.831	0.588, 1.208	0.914	0.667, 1.252
pair23-floor / pair23-crater	16/16	1.143	0.982	0.859	0.599, 1.224	0.806	0.506, 1.285

m1-ell1211653-q45
conductor gap ell=1211653 (2^20..), D=-19, hard kernel, q 45 bits: crater vs 3 floors, 128 shared targets. Record outcomes: {'verified': 1024}.

Arm `rho-neg`:

comparison	n	mean S base	mean S other	ratio	95% bootstrap	paired geo-mean	paired 95%
floor(seed1) / crater	128/128	0.905	0.918	1.015	0.881, 1.166	1.059	0.886, 1.266
floor(seed2) / crater	128/128	0.905	0.971	1.074	0.938, 1.229	1.096	0.918, 1.309
floor(seed3) / crater	128/128	0.905	0.961	1.062	0.935, 1.207	1.119	0.937, 1.335

m2-ell1138019-q45
conductor gap ell=1138019 (2^20..), D=-19, hard kernel, q 45 bits: crater vs 3 floors, 128 shared targets. Record outcomes: {'verified': 1024}.

Arm `rho-neg`:

comparison	n	mean S base	mean S other	ratio	95% bootstrap	paired geo-mean	paired 95%
floor(seed1) / crater	128/128	0.967	0.918	0.950	0.829, 1.090	0.943	0.802, 1.109
floor(seed2) / crater	128/128	0.967	0.943	0.976	0.858, 1.116	0.989	0.838, 1.166
floor(seed3) / crater	128/128	0.967	0.942	0.974	0.853, 1.115	0.995	0.857, 1.156

m3-ell1609903-q45
conductor gap ell=1609903 (2^20..), D=-19, hard kernel, q 45 bits: crater vs 3 floors, 128 shared targets. Record outcomes: {'verified': 1024}.

Arm `rho-neg`:

comparison	n	mean S base	mean S other	ratio	95% bootstrap	paired geo-mean	paired 95%
floor(seed1) / crater	128/128	0.959	0.913	0.953	0.835, 1.091	0.908	0.764, 1.079
floor(seed2) / crater	128/128	0.959	0.891	0.929	0.815, 1.060	0.936	0.807, 1.085
floor(seed3) / crater	128/128	0.959	0.881	0.919	0.810, 1.048	0.929	0.804, 1.073

s1a-ell8209-q29
conductor gap ell=8209, D=-19, hard kernel: crater vs 3 floors vs isomorphic twins, 1024 shared targets. Record outcomes: {'verified': 18432}.

Arm `rho-neg`:

comparison	n	mean S base	mean S other	ratio	95% bootstrap	paired geo-mean	paired 95%
floor1 / crater	1024/1024	0.978	0.975	0.997	0.955, 1.042	0.999	0.948, 1.053
floor2 / crater	1024/1024	0.978	0.973	0.995	0.952, 1.039	1.006	0.955, 1.060
floor3 / crater	1024/1024	0.978	0.947	0.968	0.925, 1.012	0.975	0.925, 1.027
crater-twin(u=3) / crater	1024/1024	0.978	0.952	0.973	0.930, 1.017	0.982	0.931, 1.035
floor1-twin(u=3) / crater	1024/1024	0.978	0.979	1.001	0.959, 1.045	1.017	0.966, 1.069

Arm `rho-plain`:

comparison	n	mean S base	mean S other	ratio	95% bootstrap	paired geo-mean	paired 95%
floor1 / crater	1024/1024	1.330	1.328	0.998	0.955, 1.047	0.998	0.945, 1.053
floor2 / crater	1024/1024	1.330	1.300	0.977	0.933, 1.025	0.982	0.929, 1.039
floor3 / crater	1024/1024	1.330	1.328	0.999	0.955, 1.047	1.023	0.970, 1.079
crater-twin(u=3) / crater	1024/1024	1.330	1.339	1.007	0.964, 1.053	1.012	0.958, 1.069
floor1-twin(u=3) / crater	1024/1024	1.330	1.318	0.991	0.948, 1.037	1.003	0.950, 1.060

rho-plain / rho-neg (crater, positive control, expect 1.414)	1024/1024	0.978	1.330	1.360	1.299, 1.423	1.344	1.273, 1.418

s1b-ell8209-q29
conductor gap ell=8209, D=-19, hard kernel: crater vs 3 floors vs isomorphic twins, 1024 shared targets. Record outcomes: {'verified': 18432}.

Arm `rho-neg`:

comparison	n	mean S base	mean S other	ratio	95% bootstrap	paired geo-mean	paired 95%
floor1 / crater	1024/1024	0.963	0.979	1.017	0.972, 1.062	1.020	0.968, 1.074
floor2 / crater	1024/1024	0.963	0.969	1.006	0.962, 1.052	1.000	0.949, 1.054
floor3 / crater	1024/1024	0.963	0.939	0.975	0.933, 1.019	0.969	0.920, 1.021
crater-twin(u=3) / crater	1024/1024	0.963	0.962	0.999	0.955, 1.045	0.999	0.948, 1.054
floor1-twin(u=3) / crater	1024/1024	0.963	0.993	1.031	0.987, 1.077	1.032	0.979, 1.087

Arm `rho-plain`:

comparison	n	mean S base	mean S other	ratio	95% bootstrap	paired geo-mean	paired 95%
floor1 / crater	1024/1024	1.321	1.326	1.004	0.961, 1.050	0.995	0.944, 1.047
floor2 / crater	1024/1024	1.321	1.357	1.027	0.982, 1.073	1.010	0.959, 1.063
floor3 / crater	1024/1024	1.321	1.332	1.008	0.965, 1.053	0.982	0.933, 1.035
crater-twin(u=3) / crater	1024/1024	1.321	1.351	1.022	0.978, 1.067	1.006	0.955, 1.060
floor1-twin(u=3) / crater	1024/1024	1.321	1.320	0.999	0.956, 1.044	0.991	0.942, 1.043
rho-plain / rho-neg (crater, positive control, expect 1.414)	1024/1024	0.963	1.321	1.372	1.314, 1.432	1.388	1.319, 1.461

s1c-ell8209-q29
conductor gap ell=8209, D=-19, hard kernel: crater vs 3 floors vs isomorphic twins, 1024 shared targets. Record outcomes: {'verified': 18432}.

Arm `rho-neg`:

comparison	n	mean S base	mean S other	ratio	95% bootstrap	paired geo-mean	paired 95%
floor1 / crater	1024/1024	0.967	0.973	1.006	0.963, 1.051	1.004	0.953, 1.057
floor2 / crater	1024/1024	0.967	0.980	1.013	0.968, 1.059	1.019	0.970, 1.069
floor3 / crater	1024/1024	0.967	0.971	1.005	0.963, 1.050	1.006	0.955, 1.059
crater-twin(u=3) / crater	1024/1024	0.967	0.964	0.997	0.955, 1.041	0.982	0.933, 1.034
floor1-twin(u=3) / crater	1024/1024	0.967	0.936	0.968	0.926, 1.013	0.960	0.913, 1.010

Arm `rho-plain`:

comparison	n	mean S base	mean S other	ratio	95% bootstrap	paired geo-mean	paired 95%
floor1 / crater	1024/1024	1.377	1.318	0.957	0.916, 0.999	0.955	0.906, 1.007
floor2 / crater	1024/1024	1.377	1.324	0.961	0.921, 1.005	0.946	0.897, 0.998
floor3 / crater	1024/1024	1.377	1.369	0.994	0.955, 1.037	0.991	0.939, 1.045
crater-twin(u=3) / crater	1024/1024	1.377	1.303	0.946	0.905, 0.989	0.941	0.893, 0.990
floor1-twin(u=3) / crater	1024/1024	1.377	1.339	0.972	0.931, 1.015	0.964	0.915, 1.016
rho-plain / rho-neg (crater, positive control, expect 1.414)	1024/1024	0.967	1.377	1.424	1.363, 1.487	1.420	1.348, 1.495

s1d-ell8209-q29
conductor gap ell=8209, D=-19, hard kernel: crater vs 3 floors vs isomorphic twins, 1024 shared targets. Record outcomes: {'verified': 18432}.

Arm `rho-neg`:

comparison	n	mean S base	mean S other	ratio	95% bootstrap	paired geo-mean	paired 95%
floor1 / crater	1024/1024	0.969	0.952	0.982	0.939, 1.027	0.982	0.931, 1.036
floor2 / crater	1024/1024	0.969	0.982	1.013	0.970, 1.058	1.014	0.963, 1.068

floor3 / crater	1024/1024	0.969	0.963	0.994	0.951, 1.038	1.003	0.952, 1.057
crater-twin(u=3) / crater	1024/1024	0.969	1.000	1.032	0.989, 1.078	1.045	0.992, 1.100
floor1-twin(u=3) / crater	1024/1024	0.969	0.972	1.002	0.960, 1.047	1.001	0.949, 1.056

Arm `rho-plain`:

comparison	n	mean S base	mean S other	ratio	95% bootstrap	paired geo-mean	paired 95%
floor1 / crater	1024/1024	1.351	1.314	0.973	0.934, 1.017	0.979	0.929, 1.033
floor2 / crater	1024/1024	1.351	1.325	0.981	0.940, 1.024	1.004	0.954, 1.057
floor3 / crater	1024/1024	1.351	1.306	0.967	0.925, 1.009	0.972	0.921, 1.025
crater-twin(u=3) / crater	1024/1024	1.351	1.326	0.982	0.939, 1.025	0.990	0.937, 1.045
floor1-twin(u=3) / crater	1024/1024	1.351	1.319	0.976	0.935, 1.021	0.989	0.938, 1.042
rho-plain / rho-neg (crater, positive control, expect 1.414)	1024/1024	0.969	1.351	1.393	1.334, 1.452	1.395	1.324, 1.471

s2-ell131101-q37

conductor gap ell=131101, D=-19, hard kernel: crater vs 3 floors vs isomorphic twins, 1024 shared targets. Record outcomes: {'verified': 12288}.

Arm `rho-neg`:

comparison	n	mean S base	mean S other	ratio	95% bootstrap	paired geo-mean	paired 95%
floor1 / crater	1024/1024	0.914	0.919	1.005	0.961, 1.051	0.992	0.939, 1.048
crater-twin(u=3) / crater	1024/1024	0.914	0.950	1.040	0.994, 1.086	1.034	0.979, 1.091
floor1-twin(u=3) / crater	1024/1024	0.914	0.922	1.009	0.963, 1.059	0.987	0.934, 1.043

Arm `rho-plain`:

comparison	n	mean S base	mean S other	ratio	95% bootstrap	paired geo-mean	paired 95%
floor1 / crater	1024/1024	1.322	1.313	0.994	0.950, 1.038	0.997	0.947, 1.050
crater-twin(u=3) / crater	1024/1024	1.322	1.309	0.990	0.948, 1.036	0.981	0.928, 1.037
floor1-twin(u=3) / crater	1024/1024	1.322	1.304	0.987	0.944, 1.033	0.988	0.935, 1.043
rho-plain / rho-neg (crater, positive control, expect 1.414)	1024/1024	0.914	1.322	1.446	1.383, 1.510	1.441	1.367, 1.520

t1-pairs00-02

conductor gap ell 2^25, D=-19, 53-bit, pairs 0..2: crater, crater twin, floor, floor twin (u=5), 16 shared targets. Record outcomes: {'verified': 192}.

Arm `rho-neg`:

comparison	n	mean S base	mean S other	ratio	95% bootstrap	paired geo-mean	paired 95%
pair0-crater-twin(u=5) / pair0-crater	16/16	1.005	0.882	0.878	0.629, 1.260	0.999	0.629, 1.586
pair0-floor / pair0-crater	16/16	1.005	0.943	0.939	0.616, 1.428	0.953	0.563, 1.612
pair0-floor-twin(u=5) / pair0-crater	16/16	1.005	1.078	1.073	0.740, 1.580	1.095	0.714, 1.680
pair1-crater-twin(u=5) / pair1-crater	16/16	0.905	0.874	0.965	0.653, 1.416	0.951	0.544, 1.660
pair1-floor / pair1-crater	16/16	0.905	1.002	1.107	0.739, 1.617	1.073	0.610, 1.889
pair1-floor-twin(u=5) / pair1-crater	16/16	0.905	1.230	1.358	1.054, 1.840	1.582	1.139, 2.195
pair2-crater-twin(u=5) / pair2-crater	16/16	0.868	0.942	1.086	0.748, 1.557	1.048	0.752, 1.460
pair2-floor / pair2-crater	16/16	0.868	0.761	0.877	0.637, 1.214	0.857	0.607, 1.209
pair2-floor-twin(u=5) / pair2-crater	16/16	0.868	1.086	1.252	0.907, 1.743	1.261	0.809, 1.965

t2-pairs03-05

conductor gap ell 2^25, D=-19, 53-bit, pairs 3..5: crater, crater twin, floor, floor twin (u=5), 16 shared targets. Record outcomes: {'verified': 192}.

Arm `rho-neg`:

comparison	n	mean S base	mean S other	ratio	95% bootstrap	paired geo-mean	paired 95%
pair3-crater-twin(u=5) / pair3-crater	16/16	0.876	0.977	1.115	0.723, 1.672	1.102	0.727, 1.672

pair3-floor / pair3-crater	16/16	0.876	0.940	1.073	0.760, 1.563	1.226	0.754, 1.994
pair3-floor-twin(u=5) / pair3-crater	16/16	0.876	0.926	1.057	0.715, 1.569	1.088	0.684, 1.731
pair4-crater-twin(u=5) / pair4-crater	16/16	0.992	1.141	1.150	0.823, 1.667	1.312	0.865, 1.989
pair4-floor / pair4-crater	16/16	0.992	0.689	0.695	0.488, 1.008	0.749	0.442, 1.267
pair4-floor-twin(u=5) / pair4-crater	16/16	0.992	1.149	1.158	0.789, 1.732	1.235	0.808, 1.886
pair5-crater-twin(u=5) / pair5-crater	16/16	0.851	1.030	1.210	0.883, 1.692	1.257	0.864, 1.830
pair5-floor / pair5-crater	16/16	0.851	1.072	1.260	0.900, 1.756	1.319	1.023, 1.700
pair5-floor-twin(u=5) / pair5-crater	16/16	0.851	1.091	1.281	0.921, 1.785	1.279	0.860, 1.902

t3-pairs06-08

conductor gap ell 2^25, D=-19, 53-bit, pairs 6..8: crater, crater twin, floor, floor twin (u=5), 16 shared targets. Record outcomes: {'verified': 192}.

Arm `rho-neg`:

comparison	n	mean S base	mean S other	ratio	95% bootstrap	paired geo-mean	paired 95%
pair6-crater-twin(u=5) / pair6-crater	16/16	1.083	0.980	0.905	0.637, 1.273	0.879	0.586, 1.321
pair6-floor / pair6-crater	16/16	1.083	0.980	0.905	0.623, 1.315	0.847	0.543, 1.323
pair6-floor-twin(u=5) / pair6-crater	16/16	1.083	0.952	0.879	0.625, 1.229	0.835	0.503, 1.384
pair7-crater-twin(u=5) / pair7-crater	16/16	0.890	1.085	1.219	0.865, 1.774	1.337	0.767, 2.331
pair7-floor / pair7-crater	16/16	0.890	0.970	1.090	0.755, 1.594	1.161	0.734, 1.835
pair7-floor-twin(u=5) / pair7-crater	16/16	0.890	0.953	1.071	0.752, 1.559	1.127	0.608, 2.090
pair8-crater-twin(u=5) / pair8-crater	16/16	0.792	0.941	1.189	0.810, 1.776	1.170	0.735, 1.864
pair8-floor / pair8-crater	16/16	0.792	0.821	1.037	0.698, 1.527	0.976	0.699, 1.363
pair8-floor-twin(u=5) / pair8-crater	16/16	0.792	0.847	1.069	0.695, 1.592	0.925	0.540, 1.583

t4-pairs09-11

conductor gap ell 2^25, D=-19, 53-bit, pairs 9..11: crater, crater twin, floor, floor twin (u=5), 16 shared targets. Record outcomes: {'verified': 192}.

Arm `rho-neg`:

comparison	n	mean S base	mean S other	ratio	95% bootstrap	paired geo-mean	paired 95%
pair9-crater-twin(u=5) / pair9-crater	16/16	0.950	0.977	1.028	0.737, 1.429	1.029	0.720, 1.472
pair9-floor / pair9-crater	16/16	0.950	1.070	1.126	0.818, 1.544	1.147	0.866, 1.521
pair9-floor-twin(u=5) / pair9-crater	16/16	0.950	1.029	1.084	0.808, 1.470	1.114	0.769, 1.613
pair10-crater-twin(u=5) / pair10-crater	16/16	1.135	0.900	0.793	0.578, 1.087	0.788	0.531, 1.168
pair10-floor / pair10-crater	16/16	1.135	0.990	0.872	0.641, 1.170	0.875	0.650, 1.178
pair10-floor-twin(u=5) / pair10-crater	16/16	1.135	0.956	0.842	0.631, 1.122	0.866	0.647, 1.161
pair11-crater-twin(u=5) / pair11-crater	16/16	1.147	0.894	0.779	0.598, 1.041	0.811	0.615, 1.069
pair11-floor / pair11-crater	16/16	1.147	0.961	0.838	0.613, 1.154	0.823	0.534, 1.268
pair11-floor-twin(u=5) / pair11-crater	16/16	1.147	0.796	0.694	0.455, 1.013	0.586	0.360, 0.954

t5-pairs12-14

conductor gap ell 2^25, D=-19, 53-bit, pairs 12..14: crater, crater twin, floor, floor twin (u=5), 16 shared targets. Record outcomes: {'verified': 192}.

Arm `rho-neg`:

comparison	n	mean S base	mean S other	ratio	95% bootstrap	paired geo-mean	paired 95%
pair12-crater-twin(u=5) / pair12-crater	16/16	0.762	1.090	1.431	0.977, 2.067	1.386	0.869, 2.211
pair12-floor / pair12-crater	16/16	0.762	0.774	1.016	0.692, 1.447	0.979	0.682, 1.404
pair12-floor-twin(u=5) / pair12-crater	16/16	0.762	0.781	1.025	0.664, 1.509	0.910	0.593, 1.397
pair13-crater-twin(u=5) / pair13-crater	16/16	1.057	0.847	0.801	0.571, 1.120	0.801	0.545, 1.179
pair13-floor / pair13-crater	16/16	1.057	0.748	0.707	0.462, 1.056	0.680	0.497, 0.930
pair13-floor-twin(u=5) / pair13-crater	16/16	1.057	0.861	0.814	0.558, 1.175	0.785	0.523, 1.179
pair14-crater-twin(u=5) / pair14-crater	16/16	1.003	0.935	0.932	0.682, 1.229	0.779	0.461, 1.316

pair14-floor / pair14-crater	16/16	1.003	1.025	1.022	0.705, 1.399	0.866	0.537, 1.395
pair14-floor-twin(u=5) / pair14-crater	16/16	1.003	0.764	0.762	0.536, 1.059	0.671	0.471, 0.954

t6-pairs15-17

conductor gap ell 2^25, D=-19, 53-bit, pairs 15..17: crater, crater twin, floor, floor twin (u=5), 16 shared targets. Record outcomes: {'verified': 192}.

Arm `rho-neg`:

comparison	n	mean S base	mean S other	ratio	95% bootstrap	paired geo-mean	paired 95%
pair15-crater-twin(u=5) / pair15-crater	16/16	0.878	0.954	1.087	0.733, 1.614	1.373	0.635, 2.970
pair15-floor / pair15-crater	16/16	0.878	1.146	1.306	0.901, 1.947	1.663	0.757, 3.652
pair15-floor-twin(u=5) / pair15-crater	16/16	0.878	1.080	1.230	0.853, 1.812	1.611	0.744, 3.488
pair16-crater-twin(u=5) / pair16-crater	16/16	1.062	0.887	0.835	0.554, 1.197	0.726	0.477, 1.103
pair16-floor / pair16-crater	16/16	1.062	0.761	0.716	0.487, 1.023	0.662	0.463, 0.947
pair16-floor-twin(u=5) / pair16-crater	16/16	1.062	1.023	0.963	0.699, 1.294	0.923	0.668, 1.275
pair17-crater-twin(u=5) / pair17-crater	16/16	0.895	1.050	1.173	0.845, 1.607	1.122	0.749, 1.679
pair17-floor / pair17-crater	16/16	0.895	0.868	0.969	0.679, 1.358	0.847	0.534, 1.344
pair17-floor-twin(u=5) / pair17-crater	16/16	0.895	1.026	1.146	0.829, 1.572	1.114	0.786, 1.579

t7-pairs18-20

conductor gap ell 2^25, D=-19, 53-bit, pairs 18..20: crater, crater twin, floor, floor twin (u=5), 16 shared targets. Record outcomes: {'verified': 192}.

Arm `rho-neg`:

comparison	n	mean S base	mean S other	ratio	95% bootstrap	paired geo-mean	paired 95%
pair18-crater-twin(u=5) / pair18-crater	16/16	1.076	0.977	0.908	0.635, 1.284	0.842	0.592, 1.197
pair18-floor / pair18-crater	16/16	1.076	0.854	0.794	0.562, 1.064	0.729	0.472, 1.125
pair18-floor-twin(u=5) / pair18-crater	16/16	1.076	1.181	1.097	0.798, 1.497	0.997	0.664, 1.497
pair19-crater-twin(u=5) / pair19-crater	16/16	0.930	0.963	1.035	0.715, 1.561	1.260	0.863, 1.841
pair19-floor / pair19-crater	16/16	0.930	0.991	1.065	0.702, 1.669	1.218	0.760, 1.953
pair19-floor-twin(u=5) / pair19-crater	16/16	0.930	0.925	0.994	0.658, 1.540	1.112	0.770, 1.606
pair20-crater-twin(u=5) / pair20-crater	16/16	1.258	0.851	0.676	0.469, 0.991	0.663	0.431, 1.020
pair20-floor / pair20-crater	16/16	1.258	0.858	0.682	0.487, 0.958	0.682	0.462, 1.006
pair20-floor-twin(u=5) / pair20-crater	16/16	1.258	1.129	0.897	0.637, 1.249	0.881	0.525, 1.479

t8-pairs21-23

conductor gap ell 2^25, D=-19, 53-bit, pairs 21..23: crater, crater twin, floor, floor twin (u=5), 16 shared targets. Record outcomes: {'verified': 192}.

Arm `rho-neg`:

comparison	n	mean S base	mean S other	ratio	95% bootstrap	paired geo-mean	paired 95%
pair21-crater-twin(u=5) / pair21-crater	16/16	1.188	0.798	0.672	0.469, 0.929	0.632	0.377, 1.059
pair21-floor / pair21-crater	16/16	1.188	1.133	0.954	0.666, 1.339	0.873	0.565, 1.348
pair21-floor-twin(u=5) / pair21-crater	16/16	1.188	1.211	1.019	0.754, 1.375	1.038	0.759, 1.419
pair22-crater-twin(u=5) / pair22-crater	16/16	1.095	1.004	0.917	0.659, 1.308	1.058	0.636, 1.758
pair22-floor / pair22-crater	16/16	1.095	1.093	0.999	0.707, 1.439	1.065	0.552, 2.052
pair22-floor-twin(u=5) / pair22-crater	16/16	1.095	0.981	0.896	0.619, 1.330	0.942	0.509, 1.744
pair23-crater-twin(u=5) / pair23-crater	16/16	1.128	1.070	0.948	0.666, 1.328	0.859	0.481, 1.531
pair23-floor / pair23-crater	16/16	1.128	1.073	0.951	0.696, 1.325	0.968	0.719, 1.303
pair23-floor-twin(u=5) / pair23-crater	16/16	1.128	0.919	0.815	0.565, 1.157	0.769	0.470, 1.258

Audits

session	ok	replays	receipt sha256
l1-pairs00-03	True (0 problems, 192 verified)	24/24 identical	eeee9d305271a6ac…

l2-pairs04-07	True (0 problems, 192 verified)	24/24 identical	63df9c5b03b3fa3b…
l3-pairs08-15	True (0 problems, 256 verified)	24/24 identical	ae924832bb99daef…
l4-pairs16-23	True (0 problems, 256 verified)	24/24 identical	38b964cd30fecb31…
l5r-pairs00-03	True (0 problems, 192 verified)	24/24 identical	3c9088a925c3dd28…
l6r-pairs04-07	True (0 problems, 192 verified)	24/24 identical	255c2f82dc580467…
l7r-pairs08-15	True (0 problems, 256 verified)	24/24 identical	1a585de883f2edef…
l8r-pairs16-23	True (0 problems, 256 verified)	24/24 identical	42041d1f7764ce88…
m1-ell1211653-q45	True (0 problems, 1024 verified)	1024/1024 identical	29da8815787c1a0e…
m2-ell1138019-q45	True (0 problems, 1024 verified)	1024/1024 identical	f12ca5f9177d4f04…
m3-ell1609903-q45	True (0 problems, 1024 verified)	1024/1024 identical	f3d51c7d19ce198a…
s1a-ell8209-q29	True (0 problems, 18432 verified)	18432/18432 identical	3d269081dcb8522a…
s1b-ell8209-q29	True (0 problems, 18432 verified)	18432/18432 identical	aee3f976a7de24a6…
s1c-ell8209-q29	True (0 problems, 18432 verified)	18432/18432 identical	609df87ca96578b1…
s1d-ell8209-q29	True (0 problems, 18432 verified)	18432/18432 identical	d066ffda5361fba6…
s2-ell131101-q37	True (0 problems, 12288 verified)	12288/12288 identical	0448f5176a4034f3…
t1-pairs00-02	True (0 problems, 192 verified)	24/24 identical	21224a23cb7a4f51…
t2-pairs03-05	True (0 problems, 192 verified)	24/24 identical	625ebcf7f75ad712…
t3-pairs06-08	True (0 problems, 192 verified)	24/24 identical	02d7c366a6180ffc…
t4-pairs09-11	True (0 problems, 192 verified)	24/24 identical	8db8f383a18d1424…
t5-pairs12-14	True (0 problems, 192 verified)	24/24 identical	da5b91f7016278a7…
t6-pairs15-17	True (0 problems, 192 verified)	24/24 identical	b4225928c95c9260…
t7-pairs18-20	True (0 problems, 192 verified)	24/24 identical	5fbe19c56bcd715c…
t8-pairs21-23	True (0 problems, 192 verified)	24/24 identical	b31b6289abd0b074…

Formal companion

See `lean/RepresentationHashInvariance.lean` and `lean/README.md`: transport of an r-adding walk along a group isomorphism preserves the ρ-length and is a bijection of the walk space (axioms: propext, Quot.sound).